COSC 466/566 Spring 2026
Software (and Web) Security

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Today's Class

- Introducing me
- Course policies and syllabus

Who Am I?

- Doowon Kim (doowon@utk.edu)
- Assistant Professor, EECS
- Ph.D. University of Maryland, College Park in 2020
- Used to live in Salt Lake City, St. Louis, and College Park
- Looking for undergraduates, interested in security
  - Primary research areas:
    - Computer security (measurements, data-driven, usable)
    - Computer networks
    - AI/ML for security/adversarial ML

How and where to find each other

- Email: doowon@utk.edu
- Office: Min Kao 345
- Office Hours: TBD
- Discord: You can see the link in Canvas.
- You can DM me over email or Discord.

Course Logistics

Course Goals

- Describe the role security plays in software (and Web) design
- Identify common classes of software (and Web) vulnerabilities
  - e.g., buffer overflow, XSS attacks, etc.
- Demonstrate the ability to reverse engineer software binaries, extracting secrets and exploiting bugs
  - From CTF challenges
- Demonstrate proficiency with ethical hacking

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Class Time

- Lectures
  - Cover the material related to software (and Web) security
  - Designed to leave time for student questions

- In-class practice
  - Solve real-world ethical hacking challenges via CTF challenges

Course Disclaimer

- **My promise to you**
- If anything on the schedule changes, especially assignments, due to whatever circumstances, you will NOT be harmed if it was outside your control (i.e., things will work in your favor w.r.t. grades)
- Examples: change to exam schedule or assignments, etc.

Course Details (1)

- COSC 466/566 Software (and Web) Security
  - Instructor: Doowon Kim
  - Course credit hours: 3.0
  - Meeting time: MWF 03:00PM -- 03:50PM
  - Place: MKB-524
  - Prerequisite(s): COSC 366
  - Office Hours: TBD
- Prereq.:
  - Security class: COSC366 (Intro to Cybersecurity)
- No material difference between 466/566.

Course Details (1) -- TAs

- Jaehwan Park: jpark127@utk.edu
  - Office Hours: Monday and Wednesday 1:50 -- 2:50 PM
  - MK339
- Lu Liu: lliu58@vols.utk.edu
  - Office Hours: Tuesday and Thursday 1:00 – 2:00 PM
  - MK339

Course Resources

- Canvas
  - Assignment submission and grades
- Discord:
- Course communication (announcements)
- Textbook: Not required

Your Grade

- CTF as assignment: 40%
  - Every week, you will be given a CTF task.
- Midterm Exam (1): 20%
- Midterm Exam (2): 20%
- Final Exam (a.k.a. midterm exam 3): 20%
- + a lot of bonus points

CTF

- Capture the Flag (CTF) in computer security is an exercise in which “flags” are secretly hidden in purposefully-vulnerable programs or websites.
- You will be given a task every week or two weeks.
- No late submission allowed.
- Every Friday or every two weeks Friday for demonstration.
  - Students who solve the CTF challenge will share the solution with classmates.

https://en.wikipedia.org/wiki/Capture_the_flag

What’s the CTF?

- Capture the Flag (CTF) in computer security is an exercise in which "flags" are secretly hidden in purposefully-vulnerable programs or websites.

Demo

- SignUp Bonus

- http://moa6.eecs.utk.edu/
- Registration code: cosc466

Demo

- Play with images

Midterm (and final) Exams

- In-class exams, but you need to bring your laptop.
- **Please come to the classroom and take the exams.**
- You will be given a couple of CTF challenges, and you need to find the flags.
  - The CTF challenges in the exams will be the same as (or very similar) you will solve in the class.
  - As long as you attend the class and know how to the CTF challenges given as assignment, you can also solve the CTF exams.
- Some multiple and open-ended questions.
- **No makeup exams given.**

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Letter Grade Distribution

- Grading
  - >= 93.00 A
  - 90.00 - 92.99 A-
  - 87.00 - 89.99 B+
  - 83.00 - 86.99 B
  - 80.00 - 82.99 B-
  - 77.00 - 79.99 C+
  - 73.00 - 76.99 C
  - 70.00 - 72.99 C-
  - <= 69.99 F
- I will curve the grades (by lowering the thresholds), depending on the circumstances.

Code of Ethics

- You commit to
  - Ethically study computer security for educational purposes
  - Refrain from using the knowledge gained to knowingly probe and attack computer security systems, unless having first received written permission from the owners or operators of those systems
  - Carefully consider ethical issues as your knowledge of computer security increases
  - Strive to formulate a personal code of ethics of the highest integrity

Code of Ethics

- Unethical practices include
  - Cracking passwords to gain unauthorized access
  - Deliberately spreading viruses or Trojan horses
  - Conducting a denial-of-service attack
  - Attempting buffer overflow attacks
  - Impersonating another person on a computer system you do not own

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Code of Ethics

- Failure to comply could result in
  - Suspension of computer privileges in the EECS department
  - Expulsion from UTK
  - Possible criminal prosecution

Code

- Failur
  - Sus
  - Exp
  - Pos

Code of Ethics

- No cheating (and plagiarism) allowed
  - You will be Immediately reported to the school
  - The school will make a decision for you.

Security Mindset

Security Mindset

- Security requires a particular mindset
  - i.e., Security Mindset
- Involves thinking about how things can be made to fail
  - E.g., attacker or criminal

Security Mindset (example)

- An automobile dealership.
- She was able to retrieve her car after service just by giving the attendant her last name.
- Now any normal car owner would be happy about how easy it was to get her car back,
- but someone with a security mindset immediately thinks: "Can I really get a car just by knowing the last name of someone whose car is being serviced?"